



Parshvanath Charitable Trust's
A. P. SHAH INSTITUTE OF TECHNOLOGY
(Approved by AICTE New Delhi & Govt. of Maharashtra, Affiliated to University of Mumbai)
(Religious Jain Minority)



Department of Computer Science & Engineering (AI & ML)

Price Sense AI

- Theme - [SDG 12, 9](#)
- Category - Software
- Group No - 15
- Group Members- Ritesh Jamdar, Sahil Birje, Shreesh Nalawade
- Guide- Dr. Bharti Khemani

Outline

- Introduction
- Literature Survey of the existing systems
- Limitations of the existing systems
- Problem statement and Objective
- Sustainable Development Goals
- Proposed System
- Experimental Setup(Details about input/ dataset, Performance evaluation parameters, Software and hardware setup)
- Implementation(Screenshots of all modules)
- Conclusion and future work
- References
- Research Paper Publication / Patent/ Copyright status

Introduction

- Online shopping platforms frequently change product prices.
- Users often miss the best time to purchase products at lower prices.
- Existing e-commerce platforms do not provide price history transparency.
- Manual price tracking is time-consuming and inefficient.
- There is a need for an automated system to track price changes and notify users.

Introduction

Objectives:

- To design a smart system for tracking product prices.
- To provide historical price analysis using visual charts.
- To notify users automatically on price drops.
- To support multiple e-commerce platforms.
- To ensure secure and personalized user experience.

Literature Survey of the existing system

Year	Author	Title	Technologies	Description
2023	Divya Khairkar et al.	E-Commerce Product Price Tracker Using Dynamic Pricing Algorithm	Web Scraping, Dynamic Pricing Algorithm, Alerts	This study proposes a dynamic pricing-based price tracker that extracts pricing information from multiple e-commerce sites and notifies users on price drops. It includes historical price data tracking and dynamic pricing analysis, though scalability and handling of JavaScript-based sites remain limited.
2024	Abhishek K S et al.	Insight Cart - A Product Analysis Website Using Web Scraping and Machine Learning	Selenium, BeautifulSoup, ML Algorithms	The work combines web scraping from several platforms with machine learning for product ranking and price trend prediction. It offers visual price analytics, yet it lacks real-time alerting and secure personalized tracking functionality to support individual user needs fully

Literature Survey of the existing system

Year	Author	Title	Tech	Description
2024	A. A. Patel et al.	Dynamic Price Monitoring in E-Commerce	Web Scraping, Data Analysis, Automation, Email	This research presents a real-time price monitoring system that automates data extraction and provides visual price trend analysis with email alerts. Although it advances automation, it faces challenges with handling dynamic content and anti-bot detection on modern e-commerce platforms.
2024	Fan Chen	Research on Real-Time E-Commerce Price Comparison System Using Python Web Scraping Technology	Python, BeautifulSoup, Scrapy, Selenium	This study explores real-time price comparison across e-commerce platforms using Python scraping libraries, evaluating different tools for handling static and dynamic content. It provides insights into efficient scraper selection, though it does not include end-user alerting or personalized tracking components.

Literature Survey of the existing system

Year	Author	Title	Tech	Description
2024	Nitya Sharma & Nishant Agnihotri	Web Scrapping Application	Web Scrapping APIs, Automation, Scheduled Jobs	This paper proposes a web scrapping based tracking system that extracts product price data from e-commerce sites and uses automation with scheduled jobs for periodic price updates. The work includes price history visualization and email alert integration, but it is limited in handling multi-platform dynamic content and lacks scalable user authentication.
2024	Arman Shaikh et al.	E-commerce Price Comparison Website Using Web Scrapping	Web Scrapping, Database Systems, Price Comparison	The system focuses on comparing product prices across different e-commerce platforms by extracting price data using web scrapping. It supports consumer decision-making through comparative visibility but does not include automated historical tracking or real-time user notifications on price changes.

Limitations of Existing Systems

- **Survey Analysis**

- Manual tracking of prices is inefficient.
- Static scrapers fail on JavaScript-heavy websites.
- No automated notifications for users.
- Limited support for multiple platforms.
- Poor data security and user personalization.

Limitations of Existing Systems

- **Problem Gaps**

- Existing price tracking systems are mostly manual or platform-specific.
- Basic scraping techniques fail on dynamic, JavaScript-based websites.
- Automated price monitoring and alerts are not reliably implemented.
- Historical price analysis is limited or unavailable.
- Lack of secure and scalable user-specific tracking solutions.

- **Our Contribution**

- DealDrop is an automated price tracking system that monitors products across multiple e-commerce platforms, stores price history, and notifies users of price drops, enabling informed purchasing decisions.

Problem statement

- Online product prices change frequently across e-commerce platforms, making it difficult for users to identify the right time to purchase. Existing tools lack reliable multi-platform tracking, automated price monitoring, and timely price drop alerts, resulting in missed savings opportunities.

Sustainable Development Goals

SDG 12 - Responsible Consumption and Production

- Helps users make informed purchasing decisions.
- Reduces unnecessary spending.

SDG 9 - Industry, Innovation and Infrastructure

- Uses modern AI-based web scraping.
- Implements scalable cloud architecture.

System Design

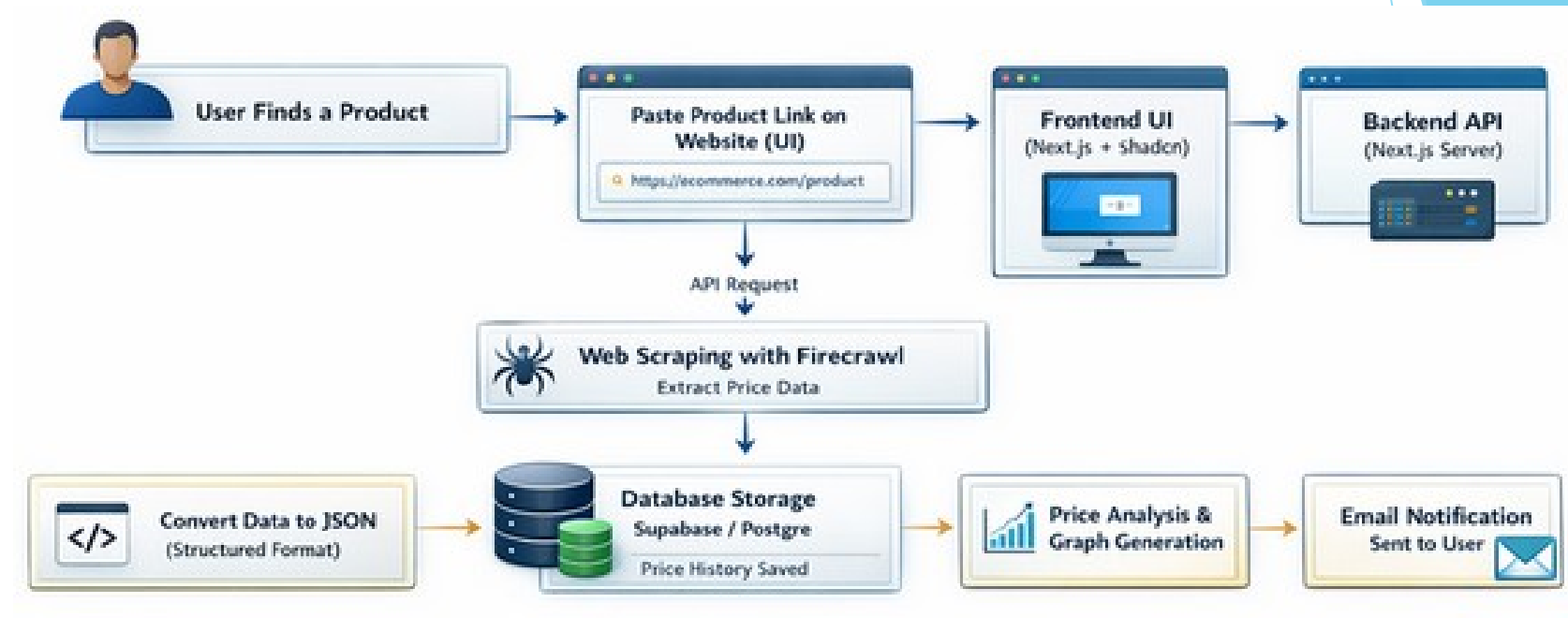


Fig 1. System Diagram

Technologies and methodologies

Frontend	Next.js, React, Tailwind CSS
Backend:	Supabase, PostgreSQL
Scraping:	Firecrawl API
Authentication	Google OAuth
Email Service:	Resend
Scheduling:	pg_cron

Implementation

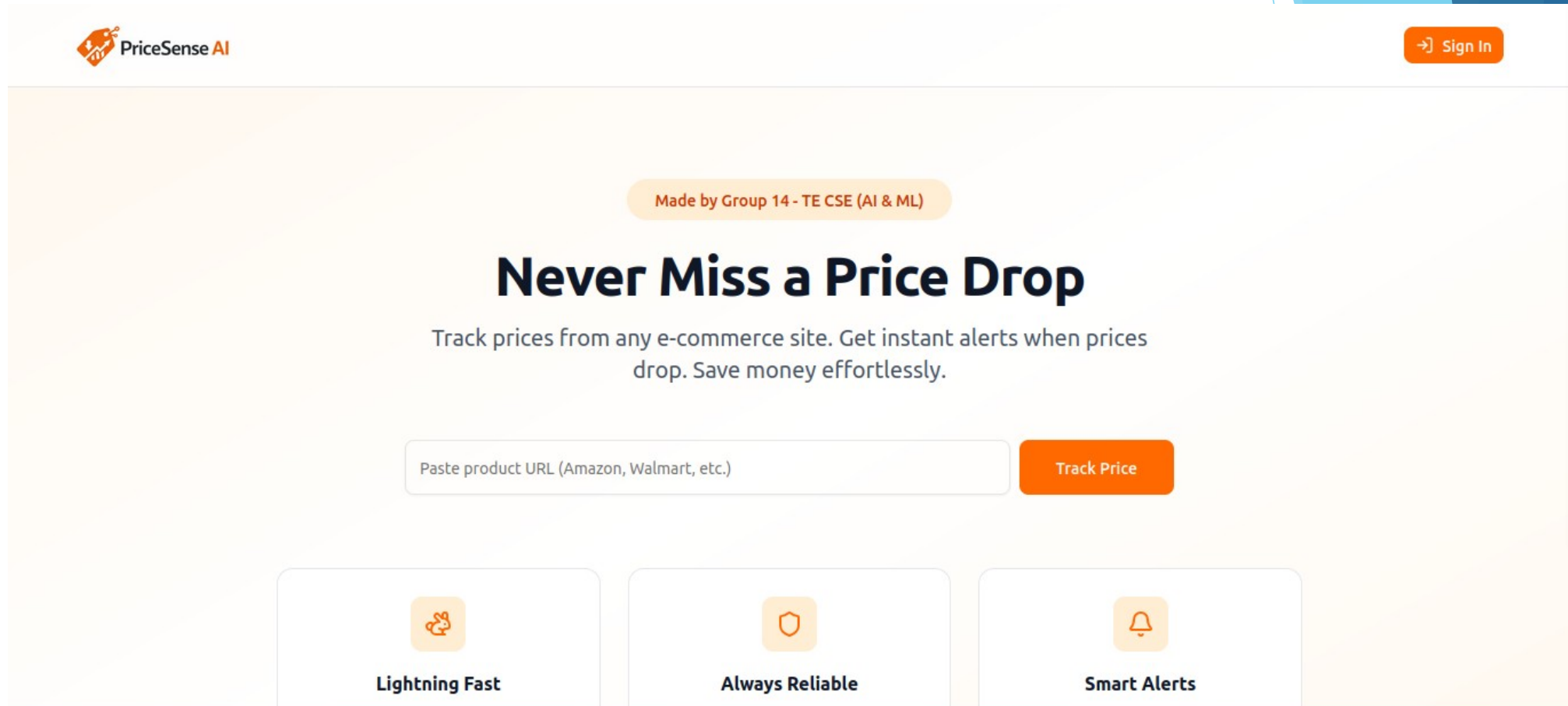


Fig 1. Landing Page

Implementation



[→ Sign In](#)

Never Miss a Price Drop

Track prices from any e-commerce site. Get instant alerts when prices drop. Save money effortlessly.

[Track Price](#)



Lightning Fast

Price Sense AI extracts prices in seconds, handling JavaScript and dynamic content



Always Reliable

Works across all major e-commerce sites with built-in anti-bot protection



Smart Alerts

Get notified instantly when prices drop below your target

Fig 2. Home Page

Implementation

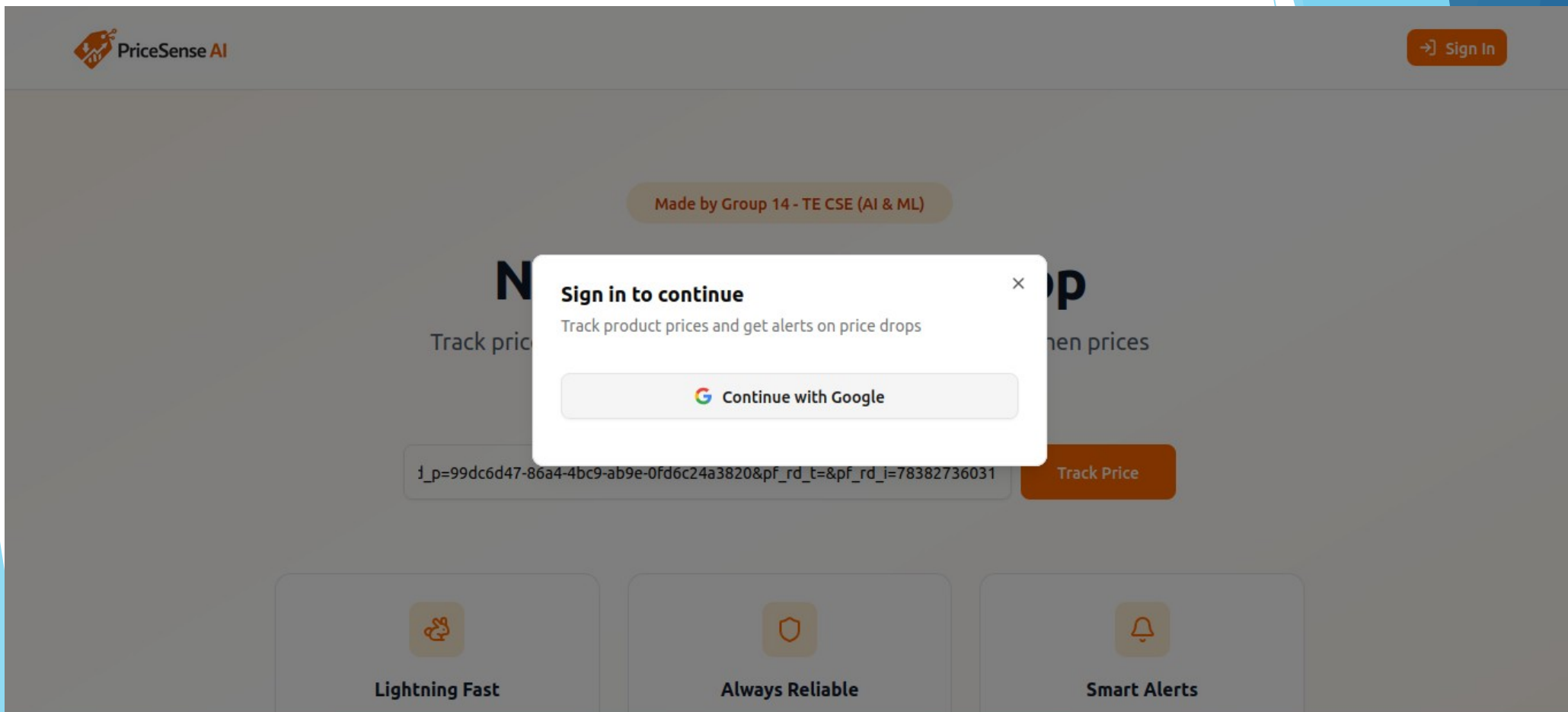


Fig 3. Sign In Page

Implementation

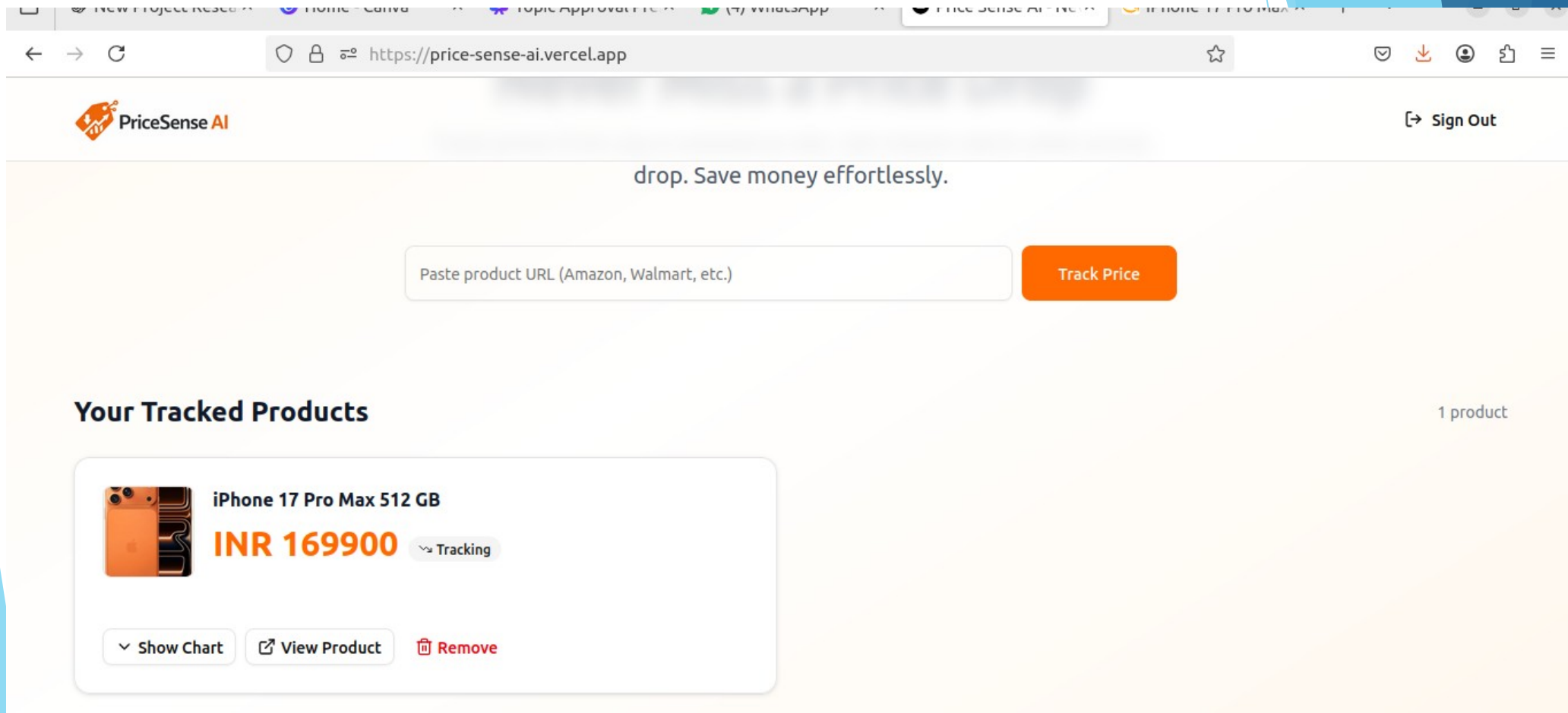


Fig 4. Price Tracking based on Product Link

References

- [1] D. Khairkar, S. Dambhare, O. Gupta, V. Thakre, N. Raut, and S. N. Dagadkar, "E-Commerce Product Price Tracker using Dynamic Pricing Algorithm," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 11, no. V, May 2023, pp. -, doi: 10.22214/ijraset.2023.51478.
- [2] A. K. S. Abhishek, S. Ravindra, B. H. Archana, H. J. Abhishek, and S. Palankar, "Insight Cart- A Product Analysis Website Using Web Scraping and Machine Learning," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 12, no. XII, Dec. 2024, pp. -, doi: 10.22214/ijraset.2024.65845.
- [3] N. Sharma and N. Agnihotri, "Web Scraping Application," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 12, no. III, Mar. 2024, pp. -, doi: 10.22214/ijraset.2024.59194.
- [4] A. A. Patel, A. R. Bhandare, A. S. Chauhan, S. V. Ghume, and T. V. Rajarshi, "Dynamic Price Monitoring in E-Commerce: Leveraging Web Scraping, Data Analysis, and Email Automation for Dynamic Pricing Insights," International Journal for Research in Applied Science & Engineering Technology (IJRASET), vol. 12, no. X, Oct. 2024, pp. -, doi: 10.22214/ijraset.2024.64786.

References

[5] “Research on Real-Time E-Commerce Price Comparison System Using Python Web Scraping Technology,” ResearchGate, 2023.

[6] “E-commerce Price Comparison Website Using Web Scraping,” ResearchGate, 2023.

Thank You...!!